Algebraic Roots & Indices

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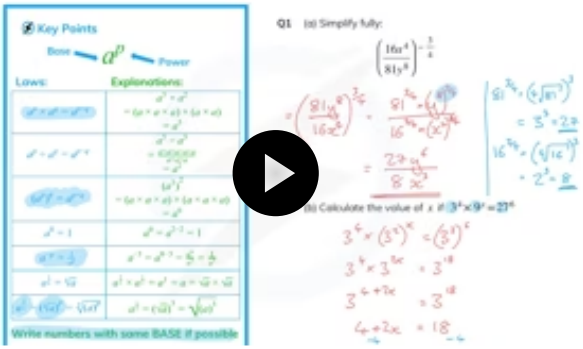

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Algebraic roots & indices

What are the laws of indices?

- **Index laws** are **rules** you can use when doing **operations** with **powers**
  - They work with both **numbers** and **algebra**

Law	Description	How it works
$a^1 = a$	Anything to the power of 1 is itself	$x^1 = x$
$a^0 = 1$	Anything to the power of 0 is 1	$b^0 = 1$
$a^m \times a^n = a^{m+n}$	To multiply indices with the same base, add their powers	$c^3 \times c^2$ $= (c \times c \times c) \times (c \times c)$ $= c^5$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

	the same base, subtract their powers	$d^4 \times d^4$ $= d^3$
$(a^m)^n = a^{mn}$	To raise indices to a new power, multiply their powers	$(e^3)^2$ $= (e \times e \times e) \times (e \times e \times e$ $= e^6$
$(ab)^n = a^n b^n$	To raise a product to a power, apply the power to both numbers, and multiply	$(f \times g)^2$ $= f^2 \times g^2$ $= f^2 g^2$
$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	To raise a fraction to a power, apply the power to both the numerator and denominator	$\left(\frac{h}{i}\right)^2 = \frac{h^2}{i^2}$
$a^{-1} = \frac{1}{a}$ $a^{-n} = \frac{1}{a^n}$	A negative power is the reciprocal	$j^{-1} = \frac{1}{j}$ $k^{-3} = \frac{1}{k^3}$
$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n}$	A fraction to a negative power, is the reciprocal of the fraction, to the positive power	$\left(\frac{l}{m}\right)^{-3} = \left(\frac{m}{l}\right)^3 = \frac{m^3}{l^3}$
$a^{\frac{1}{n}} = \sqrt[n]{a}$	The fractional power $\frac{1}{n}$ is the nth root ($\sqrt[n]{}$)	$n^{\frac{1}{2}} = \sqrt[2]{n}$ $p^{\frac{1}{3}} = \sqrt[3]{p}$
$a^{-\frac{1}{n}} = \frac{1}{a^{\frac{1}{n}}} = \frac{1}{\sqrt[n]{a}}$	A negative, fractional power is one over a root	$q^{-\frac{1}{2}} = \frac{1}{\sqrt[2]{q}}$ $r^{-\frac{1}{3}} = \frac{1}{\sqrt[3]{r}}$

Help me understand

$=(a^n)^{\frac{m}{n}}=(\sqrt[n]{a})^m$ $=(a^m)^{\frac{1}{n}}=\sqrt[n]{a^m}$	power $\frac{\ }{n}$ is the nth root all to the power m, $(\sqrt[n]{\ })^m$, or the nth root of the power m, $\sqrt[n]{(\ )^m}$ (both are the same)	$s^{\frac{2}{3}}=(s^2)^{\frac{1}{3}}=\sqrt[3]{s^2}$

- These can be used to **simplify** expressions
  - Work out the **number** and **algebra** parts **separately**
    - $(3x^7)\times(6x^4)=(3\times 6)\times(x^7\times x^4)=18x^{7+4}=18x^{11}$
    - $\frac{6x^7}{3x^4}=\frac{6}{3}\times\frac{x^7}{x^4}=2x^{7-4}=2x^3$
    - $(3x^7)^2=(3)^2\times(x^7)^2=9x^{14}$

How do I find an unknown inside a power?

- A term may have a **power involving an unknown**
  - E.g. 7^{4x}
- If both sides of an equation have the **same base number**, then the powers must be equal
  - E.g. If $4^{3x}=4^9$ then $3x=9$
  - And $x=3$
- You may have to do some **simplifying first** to reach this point
  - E.g. $3^{2x}\times 3^4=3^{18}$ simplifies to $3^{2x+4}=3^{18}$
  - Therefore $2x+4=18$
  - And $x=7$

Worked Example

(a) Simplify $(u^5)^5$

Answer:

Use $(a^m)^n=a^{mn}$

$$(u^5)^5=u^{5\times 5}$$

$$u^{25}$$

(b) If $q^x=\frac{q^2\times q^5}{q^{10}}$ find x.

Answer:

Use $a^m\times a^n=a^{m+n}$ to simplify the numerator

$$q^2\times q^5=q^{2+5}=q^7$$

$$\frac{q^7}{q^{10}} = q^{7-10} = q^{-3}$$

Write out both sides of the equation

$$q^x = q^{-3}$$

Both sides are now over the same base of q

So x must equal the power on the right-hand side

$$x = -3$$

Worked Example

(a) Rewrite $\frac{1}{\sqrt[3]{x^4}}$ in the form x^n where n is a negative fraction.

Answer:

Use $a^{\frac{1}{n}} = \sqrt[n]{a}$ to rewrite the cube-root as a power of $\frac{1}{3}$

$$\frac{1}{(x^4)^{\frac{1}{3}}}$$

Use $(a^m)^n = a^{mn}$ to simplify the denominator

$$\frac{1}{x^{\frac{4}{3}}}$$

Use $a^{-n} = \frac{1}{a^n}$ to rewrite as a term with a negative fraction as the power

$$x^{-\frac{4}{3}}$$

(b) Find the value of the constants m and a given that $(ax^6)^{\frac{1}{m}} = 8x^3$.

Answer:

Use $(ab)^n = a^n b^n$ to rewrite the left hand side
Remember to apply the power to both a and x^6

$$a^{\frac{1}{m}} \times x^{\frac{6}{m}} = 8x^3$$

Both sides of the equation have a constant part, $a^{\frac{1}{m}}$ and 8
And both sides of the equation have a part in terms of x

The two sides of the equation are equal, so set the respective parts equal to one another

First,

$$x^{\frac{6}{m}} = x^3$$

m

Solve to find m

$$m = 2$$

Then set the constant parts of both sides equal to one another

$$a^{\frac{1}{m}} = 8$$

We now know that $m = 2$, so substitute this in

$$a^{\frac{1}{2}} = 8$$

Use $a^{\frac{1}{n}} = \sqrt[n]{a}$ to rewrite as a square root

$$\sqrt[2]{a} = 8$$

Find a by squaring both sides

$$a = 64$$

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